전산화학 (2010년 1학기)

2차 시험 해답

(**총 8 문제, 65점**)

1. **(총 6 점, 각 3 점)**

(a)

$$\begin{aligned} e^{-2+i\pi/2} &= e^{-2}e^{i\pi/2} \\ &= e^{-2}\Big(\cos\Big(\frac{\pi}{2}\Big)+i\sin\Big(\frac{\pi}{2}\Big)\Big) \\ &= e^{-2}i \end{aligned}$$

$\therefore \mathrm{Re}(z)=0, \qquad \mathrm{Im}(z)=e^{-2}.$

(b) Euler 공식을 이용하면,

$$e^{i\pi/4}=\Big(\cos\Big(\frac{\pi}{4}\Big)+i\sin\Big(\frac{\pi}{4}\Big)\Big)$$

이고,

$$2e^{i\pi/2}=2\Big(\cos\Big(\frac{\pi}{2}\Big)+i\sin\Big(\frac{\pi}{2}\Big)\Big)$$

그러므로,

$$\begin{aligned} e^{i\pi/4}+2e^{i\pi/2} &= \Big(\cos\Big(\frac{\pi}{4}\Big)+i\sin\Big(\frac{\pi}{4}\Big)\Big)+2\Big(\cos\Big(\frac{\pi}{2}\Big)+i\sin\Big(\frac{\pi}{2}\Big)\Big) \\ &= \frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}i+2i=\frac{\sqrt{2}}{2}+\Big(\frac{\sqrt{2}}{2}+2\Big)i \end{aligned}$$

$\therefore \mathrm{Re}(z)=\sqrt{2}/2, \qquad \mathrm{Im}(z)=\sqrt{2}/2+2.$

2. **(총 6 점, 각 2 점)**

(a)

$$\begin{aligned} \frac{dy}{dx} &= \frac{2x\sqrt{1-5x}-x^2\frac{1}{2}\frac{-5}{\sqrt{1-5x}}}{1-5x} \\ &= \frac{4x(1-5x)+5x^2}{2(1-5x)\sqrt{1-5x}} \\ &= \frac{-15x^2+4x}{2(1-5x)^{3/2}} \end{aligned}$$

(b)

$$\frac{dy}{dx}=\frac{1}{\tan(2x)}\times\sec^2(2x)\times 2=\frac{2\sec^2(2x)}{\tan(2x)}$$

(c)

$$\frac{dy}{dx} = Ae^{-B\cos(Cx)} \times (B\sin(Cx) \times C) = ABC\sin(Cx)e^{-B\cos(Cx)}$$

3. **(총 8 점, 각 4점)**

(a)

$$\begin{aligned}
f_x(x,y) = \left(\frac{\partial f}{\partial x}\right)_y &= \frac{2abx}{bx^2+cy^2} \\
f_y(x,y) = \left(\frac{\partial f}{\partial y}\right)_x &= \frac{2acy}{bx^2+cy^2} \\
f_{xx}(x,y) = \left(\frac{\partial^2 f}{\partial x^2}\right)_y &= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)_y = \frac{2ab(bx^2+cy^2)-2abx(2bx)}{(bx^2+cy^2)^2} \\
&= \frac{2ab(-bx^2+cy^2)}{(bx^2+cy^2)^2} \\
f_{yy}(x,y) = \left(\frac{\partial^2 f}{\partial y^2}\right)_x &= \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)_x = \frac{2ac(bx^2+cy^2)-2acy(2cy)}{(bx^2+cy^2)^2} \\
&= \frac{2ac(bx^2-cy^2)}{(bx^2+cy^2)^2} \\
f_{xy}(x,y) = \left(\frac{\partial^2 f}{\partial y \partial x}\right) &= \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)_y = \frac{-2abx(2cy)}{(bx^2+cy^2)^2} \\
&= \frac{-4abcxy}{(bx^2+cy^2)^2} \\
f_{yx}(x,y) = \left(\frac{\partial^2 f}{\partial x \partial y}\right) &= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)_x = \frac{-2acy(2bx)}{(bx^2+cy^2)^2} \\
&= \frac{-4abcxy}{(bx^2+cy^2)^2}
\end{aligned}$$

(b)

$$\begin{aligned}
f_x(x,y) = \left(\frac{\partial f}{\partial x}\right)_y &= ye^x + y + \ln y \\
f_y(x,y) = \left(\frac{\partial f}{\partial y}\right)_x &= e^x + x + \frac{x}{y} \\
f_{xx}(x,y) = \left(\frac{\partial^2 f}{\partial x^2}\right)_y = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)_y &= ye^x \\
f_{yy}(x,y) = \left(\frac{\partial^2 f}{\partial y^2}\right)_x = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right)_x &= -\frac{x}{y^2} \\
f_{xy}(x,y) = \left(\frac{\partial^2 f}{\partial y \partial x}\right) = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)_y &= e^x + 1 + \frac{1}{y} \\
f_{yx}(x,y) = \left(\frac{\partial^2 f}{\partial x \partial y}\right) = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right)_x &= e^x + 1 + \frac{1}{y}
\end{aligned}$$

4. **(총 12 점, 각 3점)**

(a) $df(x,y) = 2xe^{axy}dx + 2ye^{axy}dy \Rightarrow$ 불완전미분

(b) $df(x,y) = by\cos(bx)dx + \sin(bx)dy \Rightarrow$ 완전미분 $f(x,y) = y\sin(bx) + C$

(c) $df(x,y) = x\cos y dx + y\sin x dy \Rightarrow$ 불완전 미분

(d) $df(x,y) = (x+y)^{-1}dx + (x+y)^{-1}dy \Rightarrow$ 완전미분 $f(x,y) = \ln(x+y) + C$

5. **(5점)** $f(x,y) = \sinh(ax)\cos(ay) = \frac{e^{ax}-e^{-ax}}{2}\cos(ay)$ 이므로,

$$\left(\frac{\partial f}{\partial x}\right)_y = a\frac{e^{ax}+e^{-ax}}{2}\cos(ay)$$

$$\left(\frac{\partial^2 f}{\partial x^2}\right)_y = a^2\frac{e^{ax}-e^{-ax}}{2}\cos(ay)$$

$$\left(\frac{\partial f}{\partial y}\right)_x = -a\frac{e^{ax}-e^{-ax}}{2}\sin(ay)$$

$$\left(\frac{\partial^2 f}{\partial y^2}\right)_x = -a^2\frac{e^{ax}-e^{-ax}}{2}\cos(ay)$$

그러므로

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = a^2\frac{e^{ax}-e^{-ax}}{2}\cos(ay) - a^2\frac{e^{ax}-e^{-ax}}{2}\cos(ay) = 0$$

6. **(총 8점, 각 4점)**

(a)

$$\begin{aligned}\frac{dV(r)}{dr} &= 4\epsilon\left(-12\frac{\sigma^{12}}{r^{13}} + 6\frac{\sigma^6}{r^7}\right) \\ &= -24\epsilon\frac{\sigma^6}{r^7}\left(\frac{2\sigma^6}{r^6} - 1\right)\end{aligned}$$

$dV(r)/dr = 0$ 이 되기 위해서는 위 식의 괄호 안이 0 이 되어야 하므로, $r = 2^{1/6}\sigma$ 가 되어야 한다.

(b)

$$\begin{aligned}\frac{d^2V(r)}{dr^2} &= 4\epsilon\left((-12)\times(-13)\frac{\sigma^{12}}{r^{14}} + 6\times(-7)\frac{\sigma^6}{r^8}\right) \\ &= 24\epsilon\frac{\sigma^6}{r^8}\left(\frac{26\sigma^6}{r^6} - 7\right)\end{aligned}$$

$r = 2^{1/6}\sigma$ 을 위 식에 대입하면,

$$\begin{aligned}\left.\frac{d^2V(r)}{dr^2}\right|_{r=2^{1/6}\sigma} &= 24\epsilon\frac{\sigma^6}{2^{4/3}\sigma^8}\left(\frac{26\sigma^6}{2\sigma^6} - 7\right) \\ &= \frac{24\epsilon}{2^{4/3}\sigma^2}(13-7) \\ &= \frac{144\epsilon}{2^{4/3}\sigma^2} > 0\end{aligned}$$

$dV(r)/dr > 0$ 이므로 $r = 2^{1/6}\sigma$ 에서 최소값을 갖는다.

7. **(총 8점, 각 4점)**

(a) $f(x,y)$ 가 (x^*, y^*) 에서 극값을 가지므로

$$\begin{aligned} f_x(x^*, y^*) &= 0 \Rightarrow 1 - 2x^* - y^* = 0 \Rightarrow y^* = 1 - 2x^* \\ f_y(x^*, y^*) &= 0 \Rightarrow 1 - x^* - y^* = 0 \Rightarrow y^* = 1 - x^* \end{aligned}$$

위 두 식을 연립하면 $x^* = 0,\ y^* = 1$

(b) Hessian 검사를 이용하면

$$\Delta_1 = f_{xx}(x^*, y^*) \quad \text{그리고} \quad \Delta_2 = f_{xx}(x^*, y^*) f_{yy}(x^*, y^*) - f_{xy}(x^*, y^*)^2.$$

주어진 함수 $f(x,y) = 4 + x + y - x^2 - xy - \frac{y^2}{2}$ 에 대하여,

$$\begin{aligned} f_{xx}(x,y) &= -2 \\ f_{yy}(x,y) &= -1 \\ f_{xy}(x,y) &= -1 \end{aligned}$$

그러므로

$$\Delta_1 = -2 < 0 \quad \text{and} \quad \Delta_2 = (-2) \times (-1) - (-1)^2 = 1 > 0$$

$\Delta_1 < 0$ 이고 $\Delta_2 > 0$ 이므로 $f(x,y)$ 는 $x^* = 2,\ y^* = 0$ 에서 극대값을 갖는다.

8. **(총 12점, 각 3점)**

(a) 압력과 온도에 대한 내부에너지 $U(P,T)$ 의 전미분은 다음과 같이 구할 수 있다.

$$dU = \left(\frac{\partial U}{\partial P}\right)_T dP + \left(\frac{\partial U}{\partial T}\right)_P dT. \tag{1}$$

(b) 부피와 온도에 대한 엔트로피 $U(V,T)$ 의 전미분은 다음과 같이 구할 수 있다.

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT. \tag{2}$$

(c) 압력과 온도에 대한 부피 $V(P,T)$ 의 전미분은 다음과 같이 구할 수 있다.

$$dV = \left(\frac{\partial V}{\partial P}\right)_T dP + \left(\frac{\partial V}{\partial T}\right)_P dT. \tag{3}$$

(d) (c)의 결과를 (b)에 넣에 정리하면

$$\begin{aligned} dU &= \left(\frac{\partial U}{\partial V}\right)_T \left[\left(\frac{\partial V}{\partial P}\right)_T dP + \left(\frac{\partial V}{\partial T}\right)_P dT\right] + \left(\frac{\partial U}{\partial T}\right)_V dT \\ &= \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial P}\right)_T dP + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P dT + \left(\frac{\partial U}{\partial T}\right)_V dT \\ &= \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial P}\right)_T dP + \left[\left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P + \left(\frac{\partial U}{\partial T}\right)_V\right] dT \end{aligned} \tag{4}$$

(4) 식을 (1) 식과 비교하면

$$\left(\frac{\partial U}{\partial P}\right)_T = \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial P}\right)_T$$

그리고

$$\left(\frac{\partial U}{\partial T}\right)_P = \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P + \left(\frac{\partial U}{\partial T}\right)_V$$